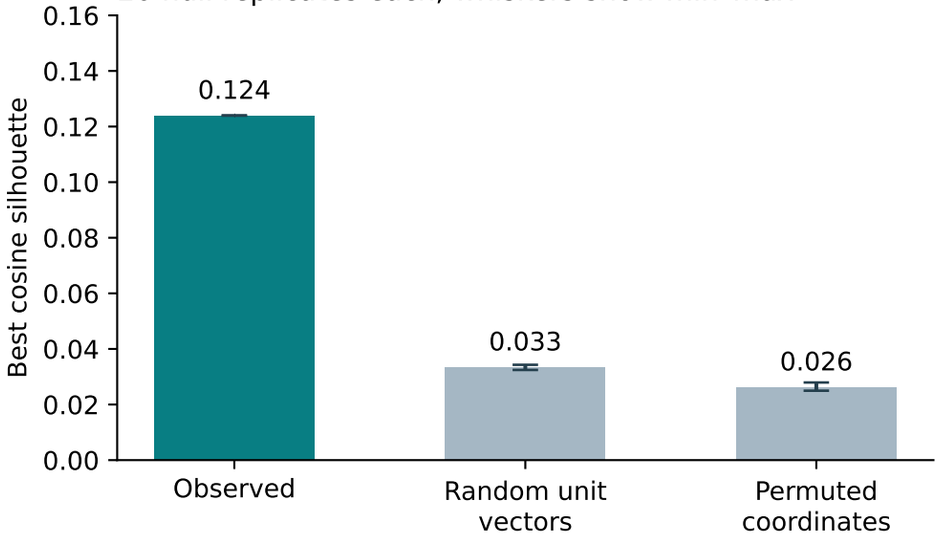
PETRI cause facets: structure, stability, and coverage

Same 801 investigations · verification ≥7 · investigator-supported statements, not verified causes

Null tests support embedding structure; stability and reading limit claims about the exact groups.

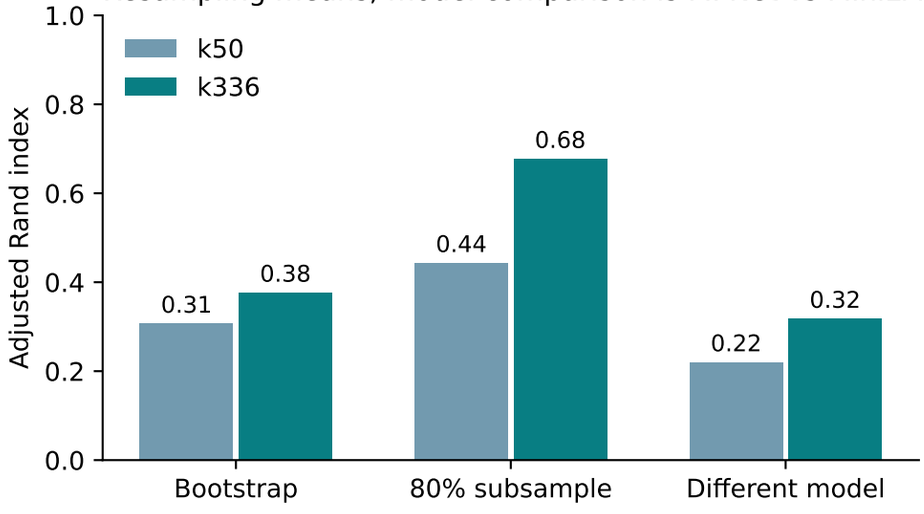
Ward: repeat the full k = 2–800 search

20 null replicates each; whiskers show min–max



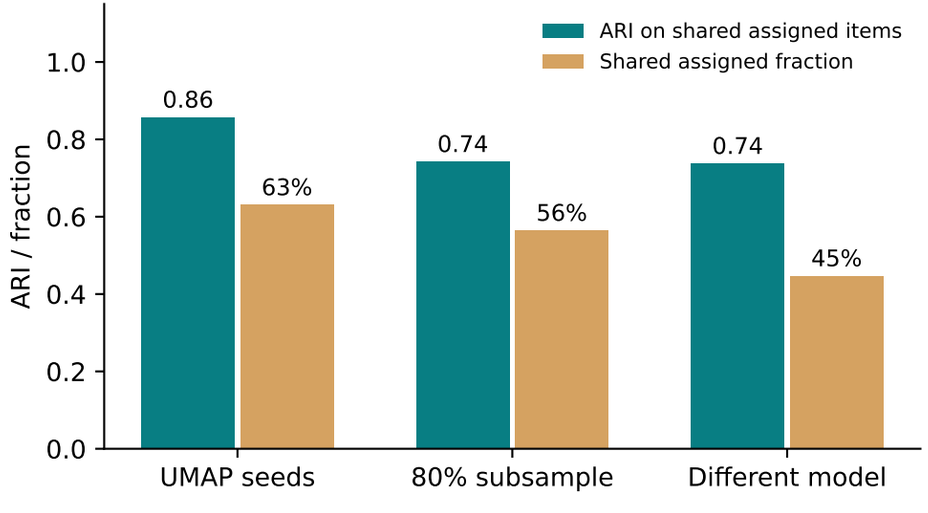
Ward: agreement with the reference partition

Resampling means; model comparison is MPNet vs MiniLM



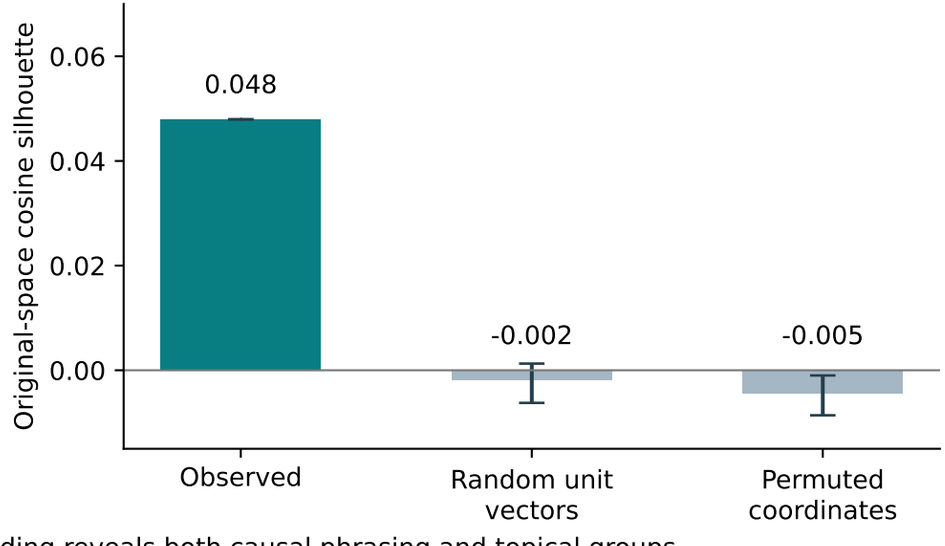
UMAP + HDBSCAN: stable core, changing coverage

Seed pairs; subsamples vs reference; model pair



UMAP + HDBSCAN: assigned records only

10 null runs each; preset settings; noise excluded



Preset HDBSCAN: 18 clusters, 563 assigned (70.3%), 238 unclustered. Reading reveals both causal phrasing and topical groups.